

Modulation of 2-Oxoglutarate Dehydrogenase Complex by Inorganic Phosphate, Mg^{2+} , and Other Effectors

José Salud Rodríguez-Zavala,¹ Juan Pablo Pardo,* and Rafael Moreno-Sánchez

Departamento de Bioquímica, Instituto Nacional de Cardiología, México, D.F., 14080, México;

**Departamento de Bioquímica, Facultad de Medicina, UNAM*

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The interplay of inorganic phosphate (Pi) with other ligands such as Mg^{2+} , ADP, ATP, and Ca^{2+} on the activation of 2-oxoglutarate dehydrogenase complex (2-OGDH) in both isolated enzyme complex and mitochondrial extracts was examined. Pi alone activated the enzyme, following biphasic kinetics with high ($K_{0.5} = 1.96 \pm 0.42$ mM) and low ($K_{0.5} = 9.8 \pm 0.4$ mM) affinity components for Pi. The activation by Pi was highly pH-dependent; it increased when the pH raised from 7.1 to 7.6, but it was negligible at pH values below 7.1. Mg-Pi and Mg-ADP, but not Mg-ATP, were more potent activators of 2-OGDH than free Pi and free ADP. ATP inhibited the 2-OGDH activity by chelating the free Mg^{2+} and also as a Mg-ATP complex. With or without Mg^{2+} , ADP, and Pi activated the 2-OGDH by increasing the affinity for 2-OG and the V_m of the reaction; ATP diminished the V_m , but it increased the affinity for 2-OG in the mitochondrial extract. Pi did not modify the 2-OGDH activation by Ca^{2+} . The results above mentioned were similar for both preparations, except for hyperbolic kinetics in the isolated enzyme and sigmoidal kinetics in the mitochondrial extracts when 2-oxoglutarate was varied. The data of this study indicated that physiological concentrations of Pi may exert a significant activation of 2-OGDH, which was potentiated by Mg^{2+} and high pH, but surpassed by ADP. © 2000 Academic Press

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2-Oxoglutarate dehydrogenase (2-OGDH) is a member of the thiamine pyrophosphate-dependent 2-oxoacid dehydrogenase complex family (1). This complex consists of three different enzymes: oxoglutarate

dehydrogenase (E1), dihydrolipoamide succinyltransferase (E2), and dihydrolipoamide dehydrogenase (E3). Functional 2-OGDH shows a stoichiometry of 12 E1, 24 E2, and 12 E3 subunits (2). The cDNA sequences of some subunits of the 2-OGDH complex from different species are already available (3–7).

Modulation of 2-OGDH activity by Ca^{2+} , ADP, ATP, GTP, NAD^+ , NADH, CoA, and succinyl-CoA is well documented (8–12). An increase in the affinity for 2-oxoglutarate is promoted by Ca^{2+} , NAD^+ , ADP, and apparently ATP (8, 10, 12). ATP as well as GTP (8) have also been described as inhibitors of the 2-OGDH activity. NADH and succinyl-CoA, the end products of the enzyme complex, induce feed-back inhibition on the first reaction (8, 10). The affinity of 2-OGDH for Ca^{2+} is enhanced by ADP and diminished by ATP (12). Mg^{2+} increased the V_m of the isolated 2-OGDH from pig heart by interacting with a binding site different to that for Ca^{2+} (13).

In a previous work we showed that changes in external Mg^{2+} , and hence in matrix-free Mg^{2+} , could modulate the 2-OGDH activity and the rate of oxidative phosphorylation in intact rat heart mitochondria (14). Mg^{2+} also affected respiratory rates in rat liver mitochondria through the modulation of the activities of the succinate and glutamate dehydrogenases (15). These observations may have physiological relevance since several recent reports have established that the cytosolic and intramitochondrial free Mg^{2+} concentrations may change in response to different hormones and agonists in different types of cells (16–20); this, however, remains controversial (21).

Mg^{2+} activation of 2-OGDH was potentiated by physiological concentrations of Pi (<5 mM), which resulted in decrease of the $K_{0.5}$ value for Mg^{2+} and elevation of the $V_{\max}$ (14). An earlier report described an activation of bovine kidney 2-OGDH by Pi, but in the concentration range of 10 to 20 mM (8), concentrations which are

¹To whom correspondence and reprint requests should be addressed. Fax: (525) 573-0926.

well above the physiological range (22). The activation by Pi was due to an increase in the affinity of 2-OGDH for 2-oxoglutarate (8).

The activation of 2-OGDH by low Pi concentrations described in intact mitochondria (14) could be the result of a direct interaction with the enzyme complex or by changing the concentration of other modulating factors such as Mg^{2+} , Ca^{2+} , or adenine nucleotides. Therefore, we examined the effect of Pi on the activity of 2-OGDH in the presence of different effectors in the purified enzyme and in mitochondrial extracts.

MATERIALS AND METHODS

Isolated pig heart 2-OGDH was purchased from Sigma Chemical Co. Two different enzyme lots were used. The enzyme solution contained 50% glycerol, 10 mg bovine serum albumin/ml, 30% sucrose, 2.5 mM EGTA, 2.5 mM EDTA, 2.5 mM 2-mercaptoethanol, 0.5% Triton X-100, 0.005% sodium azide, 25 mM potassium phosphate, pH 6.8, and 11 or 9.7 mg protein/ml; the specific activity was 0.63 or 1 units/mg protein. The enzyme solution was stored in aliquots of 1 ml at -10°C . No activity was lost after a year of storage.

Rat heart mitochondria were isolated from male Wistar rats of 250–300 g, according to a previously described method, incubating two hearts in SHE medium (250 mM sucrose, 10 mM Hepes, 1 mM EGTA, pH 7.4) in the presence of 1 mg of the protease Nagarse (protease type XXVII, Sigma Chem.; sp act 10 units/mg) on ice for 9 min (23).

Mitochondrial extracts. Freshly prepared mitochondria (5 mg protein/ml) were suspended in SHE medium that also contained 2 mM phenylmethylsulfonyl fluoride, 0.5% BSA, 2 mM EGTA, 2 mM EDTA, 1 mM dithiothreitol and solubilized with 0.2% Triton X-100. This mitochondrial solution was diluted with glycerol (1:1) and stored at -10°C until use; under these conditions the 2-OGDH activity remained constant for at least one month. After one month, activation by Pi and Mg^{2+} started to diminish, although activation by Ca^{2+} and ADP remained unchanged. In extracts prepared by freezing in liquid N_2 and fast thawing at 40°C three times (11), the 2-OGDH activity was much lower and unstable than in Triton-solubilized mitochondria.

2-OGDH activity. An aliquot of mitochondrial extract (0.5 mg protein/ml) or isolated pig heart 2-OGDH (0.055 mg protein/ml) was incubated in a medium that contained 120 mM KCl, 20 mM Mops-K (pH 7.35), 0.5 mM EGTA, 1 mM dithiothreitol, 1 mM thiamine pyrophosphate (TPP), 1 mM NAD^+ , and different concentrations of 2-oxoglutarate at 30°C . Fresh solutions of TPP, NAD^+ , and DTT were used throughout this study. Other additions were as indicated in the figure legends. The reaction was initiated 2–5 min later by the addition of 0.25 mM CoA. NADH formed was determined following the absorbance at 340 nm in a dual beam Aminco DW2000 spectrophotometer. The rate of NADH formation was linear for at least 5 min under the described conditions. The reduction of NAD^+ either in the absence of 2-oxoglutarate, CoA or TPP was negligible. The determination of 2-OGDH activity by measuring NADH absorbance instead of NADH fluorescence was used because it exhibited lower variations (25).

The concentrations of free Mg^{2+} , free Ca^{2+} (in the presence of 0.75–4.25 mM EGTA, final concentrations), Pi-Mg, ADP-Mg, and ATP-Mg were calculated with the use of the computer program CHELATOR (24). Fitting of equations to experimental data was done by using the computer program ORIGIN Ver 3.73 (Microcal Software Inc.).

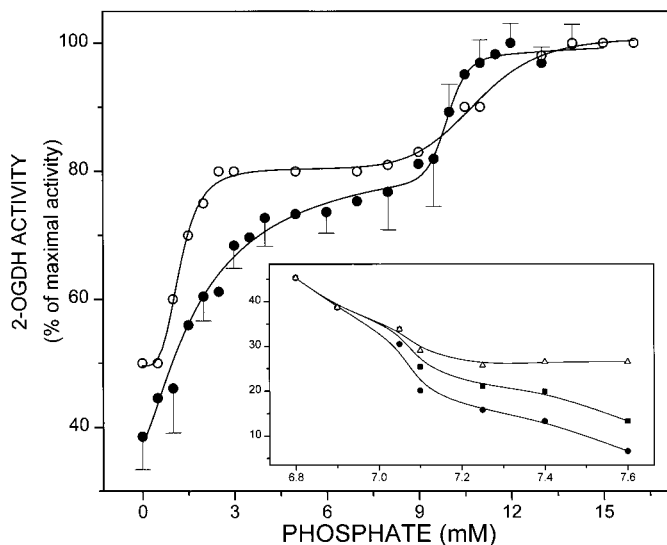


FIG. 1. Activation of 2-OGDH by Pi and pH. Aliquots of isolated enzyme (0.055 mg/ml) or mitochondrial extracts (0.5 mg/ml) were incubated in KME medium (plus 2 μM rotenone with the extracts) as described under Materials and Methods, in the presence of 0.5 mM 2-OG and the indicated Pi concentrations. $\circ$, isolated enzyme; $\bullet$, mitochondrial extracts. The pH of the reaction mixture was checked after the addition of Pi and kept constant at a value of 7.3 ± 0.05 . The solid lines represent the best fit to a double Hill equation. The values of the Hill coefficients for the isolated enzyme were 3.6 and 2.9 and for the mitochondrial extracts were 1.97 ± 0.9 and 5.4 ± 1.3 ($n = 5$) for the range of Pi concentration from 0 to 5 mM and from 5 to 15 mM, respectively. Values are mean of two experiments made with two different lots of the isolated enzyme or mean $\pm$ standard deviation of 5 different extract preparations. The maximal activity measured in the presence of 15 mM Pi was 63 ($n = 2$) and 8.3 ± 2.5 ($n = 5$) nmol NADH min^{-1} (mg protein) $^{-1}$ for the isolated enzyme and the mitochondrial extracts, respectively. Inset: the x-axis is pH, the y-axis is 2-OGDH activity in nmol NADH (mg protein) $^{-1}$ min^{-1} measured at Pi concentrations of zero (circles), 5 (squares), and 15 mM (triangles).

RESULTS

Activation by Pi and pH

To examine whether the effect of Pi on the 2-OGDH activity observed in intact mitochondria (14) was due to a direct interaction with the enzyme complex, the effect of Pi on the commercially available 2-OGDH and on extracts from rat heart mitochondria was measured. The addition of Pi induced a significant stimulation of the activity in both, the isolated enzyme and the mitochondrial extract with two kinetic components (Fig. 1): a high affinity component for Pi concentrations lower than 5 mM, and a second component of low affinity at Pi concentrations larger than 7 mM.

Variations in pH modify the 2-OGDH activity isolated from bovine kidney (8) and heart from different sources (10, 11). Therefore, the effect of pH on the activation of isolated 2-OGDH by Pi was assayed. In agreement with earlier reports (10, 11), the lowering of

TABLE I

Kinetic Parameters of the 2-OGDH Activity in the Isolated Enzyme and in Mitochondrial Extracts for Different Ligands

Ligand	Isolated Enzyme		Mitochondrial Extract	
	$K_{0.5}$ (μM)	V_m (nmol NADH/min/mg)	$K_{0.5}$ (μM)	V_m (nmol NADH/min/mg)
Pi (High affinity)	1200 (2)	49.7 (2)	1960 $\pm$ 420 (6)	6.1 $\pm$ 1.9 (6)
Pi (Low affinity)	10800 (2)	63 (2)	9790 $\pm$ 390 (4)	8.3 $\pm$ 2.5 (4)
Mg ²⁺	143.6 $\pm$ 14 (3)	58 $\pm$ 26 (3)	115 $\pm$ 30 (6)	9.5 $\pm$ 6.6 (6)
Mg-ADP	6.9 $\pm$ 2.6 (5)	238 $\pm$ 25 (5)	—	—
Mg-Pi	26 (2)	150 (2)	12 (2)	13 (2)
ADP	115 $\pm$ 35 (5)	70.5 $\pm$ 7 (5)	406 $\pm$ 112 (6)	58 $\pm$ 8 (6)
Ca ²⁺	0.224 $\pm$ 0.04 (8)	46 $\pm$ 16 (8)	0.279 (2)	52 (2)
B	K_m (mM)		$K_{0.5}$ or K_m (mM)	
2-OG	2.4 $\pm$ 1.1 (3)	138 $\pm$ 52 (3)	4.9 $\pm$ 0.5 (3)	58 $\pm$ 8 (3)
			[1.6 $\pm$ 0.5]	[64 $\pm$ 6]
2-OG (+ 0.6 mM ADP)	0.8 $\pm$ 0.2 (3)	166 $\pm$ 29 (3)	1.4 $\pm$ 0.1 (3)	72 $\pm$ 3 (3)
			[0.48 $\pm$ 0.07]	[75 $\pm$ 3]
2-OG (+ 0.6 mM ATP)	2.3 $\pm$ 0.4 (3)	105 $\pm$ 27 (3)	3 $\pm$ 0.2 (3)	34 $\pm$ 3 (3)
			[1.3 $\pm$ 0.8]	[37 $\pm$ 2]
2-OG (+ 5 mM Pi)	2.3 $\pm$ 0.8 (3)	154 $\pm$ 11 (3)	1.1 $\pm$ 0.9 (3)	74 $\pm$ 13 (3)
			[1.0 $\pm$ 0.14]	[70 $\pm$ 7]
2-OG (+ 0.6 mM Mg ²⁺)	2.6 $\pm$ 0.8 (3)	466 $\pm$ 83 (3)	5.4 $\pm$ 1.8 (4)	63 $\pm$ 3 (4)
			[1.9 $\pm$ 0.6]	[63 $\pm$ 3]
2-OG (+ 0.6 mM ADP + 0.6 mM Mg ²⁺)	1.1 $\pm$ 0.4 (3)	621 $\pm$ 48 (3)	1.2 $\pm$ 0.5 (3)	69 $\pm$ 11 (3)
			[0.48 $\pm$ 0.1]	[73 $\pm$ 13]
2-OG (+ 0.6 mM ATP + 0.6 mM Mg ²⁺)	3.9 $\pm$ 0.7 (3)	375 $\pm$ 77 (3)	2.1 $\pm$ 0.3 (3)	40 $\pm$ 12 (3)
			[0.67 $\pm$ 0.03]	[41 $\pm$ 10]
2-OG (+ 5 mM Pi + 0.6 mM Mg ²⁺)	2.2 $\pm$ 0.7 (3)	535 $\pm$ 103 (3)	1.3 $\pm$ 0.9 (4)	78 $\pm$ 11 (4)
			[0.7 $\pm$ 0.4]	[71 $\pm$ 2]

Note. Values shown are mean $\pm$ standard deviation. The ligands between parentheses were fixed at a constant concentration. Values of $K_{0.5}$ or K_m were calculated by fitting the data to the Hill or Michaelis–Menten equations. Values between square brackets represent the K_m and V_m values determined by fitting the data to the Monod–Wyman–Changeaux equation. In A, 0.5 mM 2-OG was used, while in B, 2-OG was varied from 0.1 to 15 mM.

pH from 7.6 to 6.8 promoted a sixfold stimulation of the enzyme activity. However, when the pH value was lower than 7.1, the activation by Pi disappeared (Fig. 1, inset). Enzyme activation by Pi was high at pH values between 7.2 and 7.6, which is the range of pH values determined for the mitochondrial matrix environment (26, 27). The effect of pH on the activation of the enzyme by Pi was similar when either the isolated enzyme (Fig. 1, inset) or the mitochondrial extract were used (data not shown).

Activation by Pi, Mg²⁺, and ADP

As in intact mitochondria (14), Pi potentiated the activation of isolated 2-OGDH by Mg²⁺; it produced a slight increase in the affinity for Mg²⁺, but a significant elevation of the apparent V_m (data not shown; see Table I for values). It should be noted that the $K_{0.5}$ and V_m values derived from these experiments were obtained at a nonsaturating concentration of 2-oxoglutarate; for this reason, these values should

be considered as apparent $K_{0.5}$ and V_m . ADP also increased V_m and the affinity for Mg²⁺ (data not shown), but the combination of Pi + ADP did not increase further the affinity for Mg²⁺, *albeit* it brought about a greater rise in V_m : The V_m values were 97, 216, and 275 nmol min⁻¹ mg⁻¹ for Mg-Pi, Mg-ADP, and Mg-Pi + Mg-ADP, respectively. In these experiments, the determination of the kinetic parameters was accomplished by plotting the activity against total concentrations of Mg²⁺ in the presence or absence of Pi and ADP. When the concentrations of free Mg²⁺ and Mg-complexes were considered, it was evident the following order of potency for 2-OGDH activation: Mg-ADP > Mg-Pi > Mg²⁺ (see Table I for V_m values). When the effect of Mg-ADP and Mg-Pi on the 2-OGDH activation was assayed, an additive activating effect was observed for the combination Mg-ADP + Mg-Pi. This observation suggested the existence of independent sites of binding for each complex.

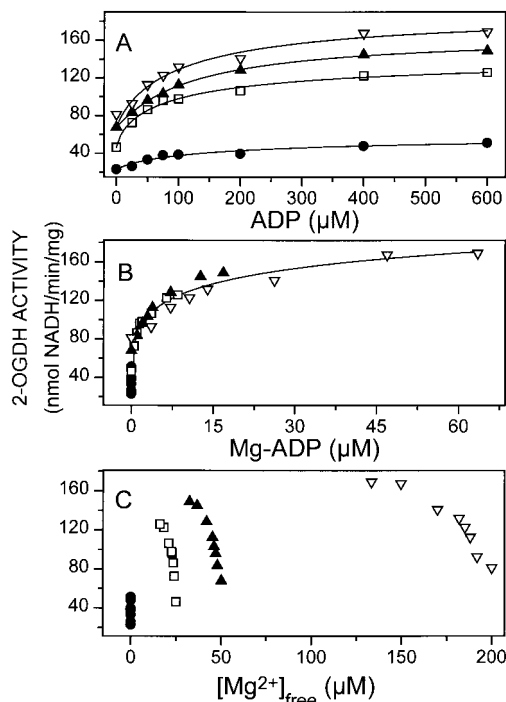


FIG. 2. Activation of isolated 2-OGDH by Mg-ADP complex. Isolated 2-OGDH (0.055 mg protein/ml) was incubated with 0.5 mM 2-OG in the following conditions: (A) ●, no Mg^{2+} added; □, +25 μM Mg^{2+} ; ▲, +50 μM Mg^{2+} ; ▽, +200 μM Mg^{2+} . (B) The Mg-ADP concentration for each point of A was calculated and plotted against activity. (C) Free Mg^{2+} concentration for each point of A was calculated and plotted against activity. Solid lines represent the best fit to the Hill equation. The Hill coefficient for curve B was not significantly different from one.

To explore further whether the Mg-ADP complex is a true activating molecule, the effect of different concentrations of Mg^{2+} and ADP on the 2-OGDH activity was assayed without (Fig. 2A) or with 5 mM Pi (data not shown). It may be noted that the activating effect of ADP in the absence of Mg^{2+} was negligible without Pi (Fig. 2A) or small with Pi (not shown). Replotting the concentrations of Mg-ADP (Fig. 2B), free Mg^{2+} (Fig. 2C), or Mg-Pi (not shown), from the data in Fig. 2A (or in the presence of 5 mM Pi), indicated that the increase in 2-OGDH activity was more closely related to variations in the Mg-ADP concentration.

Effect of ATP on the 2-OGDH Activity

In contrast to ADP, the addition of ATP induced a significant diminution of 2-OGDH activity in the presence of different concentrations of Mg^{2+} (Fig. 3A). In the absence of added Mg^{2+} (+ 0.2 mM EDTA) and with 0.5 mM 2-OG, ATP did not affect the activity of 2-OGDH (Fig. 3A). However, at higher concentrations of 2-OG, and without Mg^{2+} , ATP inhibited the 2-OGDH activity (see Figs. 4A and 5A below). The estimation of

free Mg^{2+} and Mg-ATP complex concentrations, from the data of Fig. 3A, revealed that inhibition of 2-OGDH activity by ATP was roughly related to the diminution in the concentration of free Mg^{2+} (Fig 3B), and also to the increase in the concentration of the Mg-ATP complex (Fig. 3C).

Modulation of the 2-OGDH Kinetics

To further understand the modulation of 2-OGDH by Pi and its interplay with Mg^{2+} and other effectors, the effect of ADP, ATP, and Pi with or without Mg^{2+} on the affinity of 2-OGDH for 2-OG and on the V_m was determined. In the absence of Mg^{2+} , ADP and Pi diminished the $K_{0.5}$ of 2-OGDH for 2-OG and enhanced the true V_m , while ATP induced a significant diminution on the V_m and on the affinity of 2-OGDH for its substrate (Fig. 4A) (see Table I for values of K_m and V_m). In the presence of Mg^{2+} , ADP (or rather Mg-ADP) also increased the V_m and the affinity for 2-OG, whereas ATP induced a slight diminution in the V_m and in the affinity for the substrate; Pi also increased the affinity for 2-OG and the V_m of the reaction, but to an extent lower

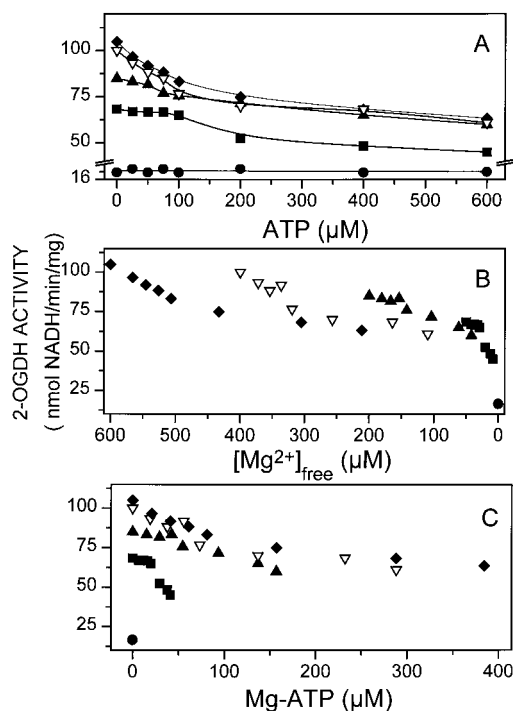


FIG. 3. Inhibition of isolated 2-OGDH by ATP. Isolated 2-OGDH (0.055 mg protein/ml) was incubated with 0.5 mM 2-OG in the following conditions: ●, no Mg^{2+} added; ■, +50 μM Mg^{2+} ; ▲, +200 μM Mg^{2+} ; ▽, +400 μM Mg^{2+} ; ◆, +600 μM Mg^{2+} . (B and C) Concentrations of free Mg^{2+} and Mg-ATP complex were calculated for each point of A and plotted against activity. Note that the X-axis in B decreases from left to right. Plots are the mean $\pm$ standard deviation of three different experiments. Solid lines represent a fit to a single exponential decay, but they have no mechanistic implications.

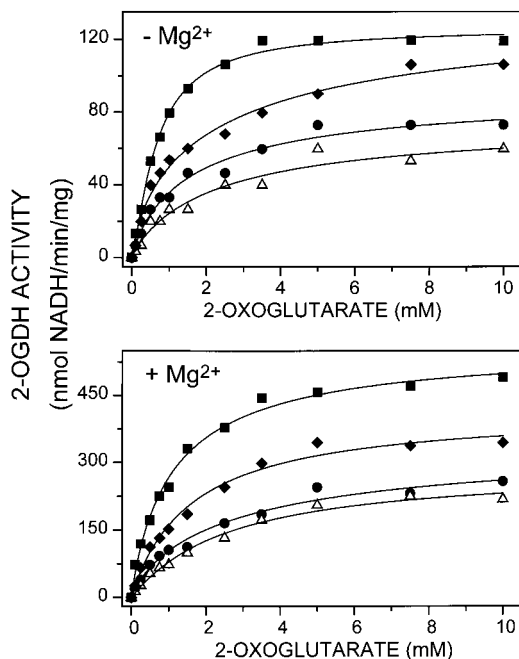


FIG. 4. Effect of ADP, ATP, Pi, and Mg^{2+} on the kinetics of the isolated 2-OGDH. Isolated 2-OGDH (0.055 mg protein/ml) was incubated with different concentrations of 2-OG. (A) ●, no further additions (control); ◆, +5 mM Pi; ■, +600 μM ADP; △, +600 μM ATP. (B) 600 μM Mg^{2+} was added. Solid lines represent the best fit to the Hill equation. Values are the mean $\pm$ standard deviation of three different experiments. The Hill coefficients values for A and B were 0.84–1.16, which were not significantly different from one indicating that the data fit well to the Michaelis–Menten equation.

than that observed with ADP (Fig. 4B). In all conditions, Mg^{2+} increased by three fold the V_m . The data of Fig. 4 indicated the following order of potency for stimulation of 2-OGDH activity: $\text{Mg-ADP} > \text{Mg-Pi} > \text{Mg}^{2+} > \text{ADP} > \text{Pi}$.

In the mitochondrial extracts, the effect of Pi, Mg^{2+} , ADP, and ATP on the affinity of 2-OGDH for 2-OG and V_m (Fig. 5) was similar to that observed in the isolated enzyme. However, in contrast to the isolated enzyme (cf. Fig. 4), mitochondrial extracts exhibited well-defined sigmoidal kinetics (Fig. 5). The Hill coefficients were not significantly different from one for the isolated enzyme, but they were higher than one ($P < 0.01$) in the mitochondrial extracts (see legends to Figs. 4 and 5 for values).

Activation by Pi, Mg^{2+} , and Ca^{2+}

To assess whether the Ca-Pi complex also influenced the activity of isolated 2-OGDH, the stimulation by Ca^{2+} at different Pi concentrations with or without ADP was measured. The results indicated that the $K_{0.5}$ value for Ca^{2+} (0.27–0.34 μM) was not modified by the presence of Pi or ADP (data not shown); similar $K_{0.5}$ values for Ca^{2+} have been previously reported by oth-

ers (12, 28, 29). Increasing Mg^{2+} in the physiological range (100–600 μM) did not alter the Ca^{2+} activation of 2-OGDH, but it promoted the elevation of the apparent V_m (data not shown). Hence, instead of an antagonistic effect between the two cations, there is a synergistic activating effect of Ca^{2+} and Mg^{2+} on 2-OGDH, through an increase in both, the affinity for 2-OG (Ca^{2+}) and the catalytic turnover (Mg^{2+}).

Table I summarizes the effect of Pi and the other effectors on the $K_{0.5}$ or K_m value for different ligands and the apparent or true V_m obtained with the isolated enzyme and the mitochondrial extracts. The $K_{0.5}$ value of the enzyme for ADP was about threefold higher in the extracts, while the $K_{0.5}$ for Pi was slightly larger than the value in the isolated enzyme. The $K_{0.5}$ value for Mg^{2+} and the K_m for 2-OG (+ ATP, + Mg^{2+} , and + Mg^{2+} + ADP) in the extracts were similar to those observed with the isolated enzyme. The $K_{0.5}$ values for Ca^{2+} and Mg-Pi and the K_m for 2-OG (in the absence of effectors, + ADP, + Pi, + ATP + Mg^{2+} , and + Pi + Mg^{2+}) were lower than those determined in the isolated enzyme.

Analysis of the V_m values revealed other differences between both preparations. In the isolated enzyme, the addition of Mg^{2+} increased 3–4 times the V_m , when the

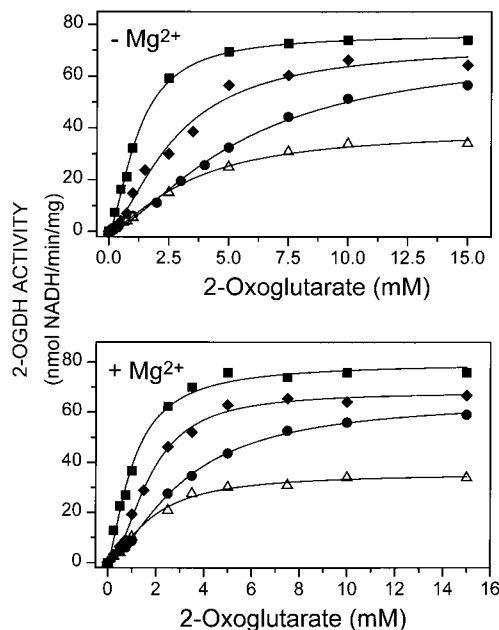


FIG. 5. Effect of ADP, ATP, Pi, and Mg^{2+} on the 2-OGDH kinetics in the mitochondrial extract. Mitochondrial extract (0.5 mg protein/ml) was incubated with different concentrations of 2-OG. (A) ●, no further additions (control); ◆, +5 mM Pi; ■, +600 μM ADP; △, +600 μM ATP. (B) 600 μM Mg^{2+} was added. Curves are representative of three different extract preparations. The Hill coefficients values for A and B ranged from 1.24 to 3.2, and they were significantly different from one ($P < 0.01$; Student *t* test for nonpaired samples). The K_m and V_m values for these plots were determined by fitting the data to the Hill and Monod–Wyman–Changeaux equations.

2-OG concentration was varied. In contrast, in the mitochondrial extracts, Mg^{2+} did not enhance further the V_m , in comparison to values obtained in its absence. The V_m values obtained with the isolated enzyme and the extracts were of a similar magnitude to others previously described (12, 13).

DISCUSSION

It was previously reported that Mg^{2+} (13, 14) and Pi (8, 9, 14) activated 2-OGDH and that Pi modulated the Mg^{2+} activation in intact heart mitochondria (14); however, the mechanism of activation by Pi was not elucidated. Here we show that interaction of Pi with 2-OGDH enhances the apparent V_m and that Mg-ADP, but not Mg-ATP, is a potent activator of the isolated 2-OGDH. The observation that Mg-Pi stimulates the 2-OGDH activity in the presence of saturating Mg-ADP, suggests the existence of different sites of modulation in the enzyme complex for these effectors. The activation of 2-OGDH by ADP has been well documented (8–11), but according to our data, the ADP site can also be occupied by the Mg-ADP complex. The previously reported activating effect of ADP (8–11) was determined in a reaction medium that contained 1 (10, 11) or 2 mM (8, 9) Mg^{2+} ; in consequence, the true activator of 2-OGDH could be Mg-ADP.

In contrast, ATP clearly exerts an inhibitory effect on 2-OGDH activity. Since Mg^{2+} induces an increase in the apparent V_m , then ATP, by decreasing the concentration of free Mg^{2+} , and as a Mg-ATP complex, brings about a diminution in the V_m . Chelation of bound Mg^{2+} by ATP might also be involved, since in the mitochondrial extracts in which added Mg^{2+} does not activate the enzyme, presumably due to the presence of bound Mg^{2+} , ATP exerts a stronger inhibition (50–60% diminution of V_m) than in the isolated enzyme (14–24% V_m diminution). However, this explanation is not compatible with the finding that addition of Mg^{2+} does not overcome the inhibitory effect of ATP (Fig. 3, Table IB). Therefore, the inhibitory effect of ATP, particularly at saturating concentrations of 2-OG and without added Mg^{2+} , seems also related to an interaction of free ATP with the enzyme.

ATP also attenuates the 2-OGDH activation by Ca^{2+} (12). However, ATP increased the affinity for 2-OG, leading to enzyme activation. Therefore, the effect of ATP on 2-OGDH activity will depend on the specific incubation conditions, although in the presence of Ca^{2+} and Mg^{2+} , a strong inhibition can be predicted, whereas in the absence of the cations a slight activation would be expected (see Figs. 3–5 and Table I).

The activation of 2-OGDH by lowering pH (Fig. 1, inset) is in agreement with previous observations in pig heart 2-OGDH (10), and bovine heart and kidney enzymes (8). The 2-OGDH activation by Pi is also pH

dependent, which is negligible at low pH and enhanced at high pH. This suggests that HPO_4^{2-} is the active Pi species or that at high pH values, some deprotonated aminoacid residues are involved in the binding of Pi. This finding may explain why the effect of Pi at physiological concentrations was not observed before and why other authors (8) observed activation by Pi solely at concentrations higher than 15 mM.

When the affinity of isolated 2-OGDH for 2-OG is determined under different conditions, the experimental curves show hyperbolic kinetics in agreement with earlier reports (9, 13), whereas with our mitochondrial extracts, the curves are clearly sigmoidal, as in toluene-permeabilized mitochondria and other mitochondrial extracts (12). Other differences between both preparations are: the negligible Mg^{2+} activation in the extracts and pronounced increase in V_m (threefold) by Mg^{2+} in the isolated enzyme; the diminution in the K_m for 2-OG induced by ADP, ATP, and Pi in the extracts, but only by ADP in the isolated enzyme; and the increase (3.5 times) in $K_{0.5}$ for ADP in the extracts at a low 2-OG concentration (see Table I).

Activation by ADP is mainly mediated by an increase in the affinity for 2-OG, as previously reported by others (8–12, 28). The stronger ADP (and Pi) activation in the extracts than in the isolated enzyme may be a consequence of the cooperativity (sigmoidal kinetics) exhibited by the former preparation. ATP increases the affinity of the enzyme for 2-OG with or without Mg^{2+} in the mitochondrial extract, but in the isolated enzyme ATP does not affect the affinity of the 2-OGDH for 2-OG.

The kinetic differences between the isolated enzyme and the mitochondrial extracts suggest that in the process of enzyme isolation, there is a change in the conformation or structure of the enzyme complex or a factor is lost, which affects cooperativity. This may occur either by the modification of the occupancy of allosteric and nonallosteric sites by different ligands or by the perturbation of protein–protein interactions in the enzyme complex. The kinetic differences appear not to be related to the presence or absence of a single ligand. Therefore, they could be due to different ligand (Mg, ADP) binding patterns between the two preparations, or to permanent structural changes such as limited proteolysis of specific subunits.

The activation of 2-OGDH by Ca^{2+} is not inhibited by increasing the Mg^{2+} concentration in the physiological range (100–600 μM), but rather the effect of both cations is additive. After agonist stimulation, $[\text{Mg}^{2+}]_c$ raises from 0.5 to 0.8 mM (16–20), $[\text{Ca}^{2+}]_c$ also increases from 0.1 to 0.6 μM (30), cytosolic Pi concentration rises from 0.83 to 3.13 mM, and the pH remains constant at 7.35 in the whole rat heart (22). These changes may produce a synergistic activating effect of 2-OGDH by both divalent cations and Pi, which may

stimulate the rate of oxidative phosphorylation (14). In consequence, the finding that Pi modulates the 2-OGDH activity may have physiological relevance.

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